

Power of their stories: faces of kernicterus – experience and perception of caregivers and their children in Northern Nigeria

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ABSTRACT

Background Our qualitative study was conducted to explore real-life experiences and challenges of caregivers and individuals with kernicterus spectrum disorder (KSD) in Northern Nigeria, where it remains a significant cause of neurological impairment. The lived experiences of affected individuals and their caregivers are largely undocumented. These stories of caregivers living with children with KSD and survivors with KSD highlight challenges and obstacles they face beyond bilirubin levels and quantitative measures.

Methods We conducted focus group discussions (FGDs; n=7), caregivers (n=54 or 6–8 per FGD) and case studies (n=4 individuals with KSD). Data were analysed using thematic analysis at three teaching hospitals in Northern Nigeria, all members of the Stop Kernicterus and Infection in Northern Nigeria+ collaborative. Caregivers with a child who had acute bilirubin encephalopathy and now has KSD and those living with KSD were interviewed.

Results Key features identified among caregivers were remarkably limited awareness of neonatal jaundice (NNJ) and its consequences, as well as reliance on traditional remedies, resulting in delays accessing care and pervasive stigma. Caregivers reported significant physical, long-term emotional and financial burdens, while individuals with KSD faced real-life challenges with mobility, communication and social inclusion. Despite these obstacles, participants demonstrated incredible resilience and inspiring aspirations.

Conclusions Our study highlights the lives and stories of caregivers and individuals affected by KSD and the urgent need for public health interventions that could prevent KSD through timely access to screening, diagnosis and treatment of NNJ. For those individuals who do progress to KSD, we need not only improved care and support for caregivers but also to value their stories, insisting they not be victimised but given every opportunity to achieve their dreams and full potential.

INTRODUCTION

Adverse outcomes of severe neonatal jaundice (NNJ)/hyperbilirubinaemia, including acute bilirubin encephalopathy (ABE) and

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ Acute bilirubin encephalopathy (ABE) from severe hyperbilirubinaemia/neonatal jaundice remains an important and preventable cause of neurological impairment in neonates and beyond kernicterus spectrum disorder (KSD).
- ⇒ Many quantitative studies exist that examine the consequences of ABE, including cerebral palsy, ABE and KSD in low- and middle-income countries (LMICs) like Nigeria, documenting general themes of prevalence but failing to depict the true narrative impact these afflictions have on families, patients and caretakers.

WHAT THIS STUDY ADDS

- ⇒ This study is the first of its kind, per an extensive literature review to qualitatively tell the stories of caregivers and children/adults living with KSD in LMICs like Nigeria, highlighting their challenges and the burden of this disease in the locales where it is most prevalent.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ The narratives shed light on their life experiences and provide much-needed evidence to promote research into KSD prevention as well as care for affected individuals and their families/caregivers, the majority of whom do not have access to the same tools, resources and technology as their counterparts in high-income countries.

kernicterus spectrum disorder (KSD), are largely preventable causes of morbidity and mortality in Nigeria and worldwide.^{1–9} Those who do survive the neonatal period often progress to KSD if hyperbilirubinaemia is untreated or treated late.^{1–3} Although ABE/KSD are not eliminated from high-income countries (HICs), severe cases are most prevalent in low- and middle-income countries

(LMICs), like Nigeria, due in part to genetic factors like glucose-6-phosphate dehydrogenase deficiency (G6PDd), weak health systems and inadequate infrastructure for appropriate diagnosis and treatment.^{4 10–14}

The occurrence of KSD is devastating to families because it is associated with various grades of disabilities, from sensorineural hearing impairment and/or language processing disorders and/or choreoathetoid cerebral palsy (CP), with significant limitations in independence and quality of life.^{11 15 16} Many affected individuals have normal intelligence, often with a brilliant mind trapped in a body that does not work or does not work well.^{3 17–20} People affected by KSD and their caregivers face multiple challenges.

Advanced technologies such as eye-gaze computers, hearing aids, electric-powered wheelchairs, feeding pumps and ramps to aid daily living activities, as well as special educational opportunities, are expensive and less available in LMICs than in HICs. They and their families often experience stigma and exclusion from school, sports, jobs and many other activities. Caregivers, largely family members, experience caregiver fatigue as they cater to the affected persons' physical, psychological, medical, educational, social and emotional needs. Providing for these needs and meeting the challenges of adults with KSD, who may outlive their parents, can be even more difficult than caring for them as children.

While the medical aspects of KSD are well-documented and quantitative studies abound, the lived experiences of those affected and their caregivers remain underexplored, particularly in LMICs, with a few studies focusing on all-cause CP and not CP from KSD.^{18 21–23} In this qualitative study, we aimed to address this gap by revealing the many faces of KSD, elucidating perspectives and experiences of caregivers of children with KSD and affected individuals themselves in Northern Nigeria.

METHODS

Study design and participants

Purposive sampling was used to select study participants through the clinics of participating institutions. Participation was voluntary and no caregivers who were invited declined participation. Our study population included caregivers with a child who had ABE and now have KSD (n=54). Additionally, we interviewed four individuals with KSD. Study sites included three teaching hospitals in Northern Nigeria, (Ahmadu Bello University Teaching Hospital (ABUTH), Zaria; Aminu Kano Teaching Hospital (AKTH), Kano and Jos University Teaching Hospital (JUTH)), Jos; all members of the Stop Kernicterus and Infection in Northern Nigeria+ collaborative.

This study employed a qualitative design to describe caregivers' perceptions of NNJ and KSD, examine their experiences in caring for persons affected by KSD and provide in-depth insight into the lived experiences of persons living with KSD. The approach enabled

comprehensive, robust and nuanced perspectives on these experiences.

We conducted seven focus group discussions (FGDs) in Hausa and/or English with caregivers (one group all males, the other six all females) and a case study of young adults with KSD, three FGDs from Kano, two each from Jos and Zaria (each FGD having 6–8 participants). Inclusion criteria were caregivers whose children had suffered severe NNJ with ABE and now have KSD and were ≥ 6 months of age, and persons with KSD. Those with KSD who could write and who consented were given a structured questionnaire to describe their experiences and challenges in their own time in writing. We developed interview guides and pilot-tested and refined them.

The FGDs were conducted at different times over an 8-week period. Focus groups were conducted in private meeting rooms within participating hospitals with only participants and the research team present. Two facilitators guided each discussion; one served as a notetaker while the other facilitated the discussion. The facilitators (n=10) included six males and four females. Seven of the facilitators were paediatricians with an average of 5 years of experience in the care of neonates and children with ABE/KSD, one each a community physician, a social scientist and an otorhinolaryngologist. Six of the facilitators had training and experience in qualitative research. The facilitators were not directly involved in the clinical management of most participants and had no prior relationship with them. The teams informed the participants about the research purpose and the professional background of the facilitators before discussions began.

We began FGDs with facilitators presenting a general overview of the group's purpose, ground rules and icebreakers. The facilitators asked open-ended questions with probes. For example, there were questions like what you knew about NNJ before your baby was affected? What was your source of information (with probes such as family members, healthcare providers)? What, in your view, are the causes of NNJ? How is NNJ treated in your community? How does having a child with KSD affect your family dynamics? What is your child's daily life like? Each focus group lasted approximately 60–90 min. Data collection continued until thematic saturation was achieved, when no new themes emerged from additional discussions. Audio-recordings of FGD and accompanying observer notes were translated and back-translated from Hausa into English and then transcribed verbatim in English by a professional company (Aminu Mende Inc, Zaria, Nigeria). Transcribed data was read multiple times to gain familiarity and to prepare for iterative analysis. Transcripts were not returned to participants for comment. Data were analysed using thematic analysis informed by the constant comparative method.²⁴ Qualitative analysis was manually performed. Two members of the research team (AA and ZLF) independently coded the transcripts and compared codes to develop a preliminary coding framework. Codes were grouped into broader categories and themes using an iterative process. Discrepancies were

resolved through discussion until consensus was reached. An ethnographic summary was done on the submissions of the persons living with KSD and relevant quotations were included.

The Consolidated Criteria for Reporting Qualitative Research (COREQ) guidelines developed by Tong *et al* were used to report this study.²⁵ We employed the COREQ checklist to ensure transparency in the reporting of research team characteristics, study design, data collection and analysis.

RESULTS

The main themes identified were: (1) awareness of NNJ; (2) knowledge of causes of NNJ; (3) treatment practices; (4) delays in accessing care; (5) caregivers' experiences with having children with KSD and (6) family and community impact, including stigma, community beliefs and misconceptions.

Awareness of NNJ

Most participants were aware of NNJ before their children developed NNJ. Sources of information included spouses, relatives, friends and healthcare providers. Examples from FGD included:

I heard that when they give birth to a child whose body suddenly becomes yellowish, he is said to have jaundice and that to treat it, the child can be brought out and kept under the sun. 'Bula' [*Prussian Blue, a blue substance used during laundry*] can also be used [*to cure jaundice*]. (FGD/female caregiver/Zaria)

I have heard of it, just that I didn't see a child with it. But I heard of jaundice. Most times, when we come for antenatal care, we heard women say they had a child with jaundice, yellowish coloration of the white of the eye. That's what I heard before the incident. (FGD/female caregiver/Kano)

There were, however, outliers (n=4) who had not heard about NNJ before their children were affected. For instance, a participant said, "I never heard about jaundice until my child was affected" (FGD/male caregiver/Jos).

Some participants mentioned spouses and cousins as some of those affected in the family. Others included friends and other relatives as those affected in the community. Some of the captured responses are reported below and others in online supplemental table 1:

I had a relative whose baby had severe jaundice, who died with CP. And her child before the index baby, died at the age of two months. We met with some others here in the clinic. (FGD/female caregiver/Zaria)

Knowledge of causes of NNJ

The participants outlined several conditions as causes of NNJ. These included illnesses during pregnancy, such as malaria and diabetes, non-compliance with routine antenatal care drugs, dietary habits during pregnancy, blood incompatibility, transmission from mother-to-baby and maternal illness (reported and in online supplemental table 1). One participant said,

I think maternal illness (malaria, for example) has a major role. As for me, I was ill from the beginning of the pregnancy till five months. I had a preterm delivery at 35 weeks+three days and the baby was born underweight, weighing 1.9 kg. Additionally, I had a retained placenta, and it took 15 hours to remove it. The baby later developed sepsis. (FGD/female caregiver/Jos)

Another said,

I don't believe that maternal illness causes jaundice because I was well throughout my pregnancy. I have four children. three were normal at birth. Even though this one was normal at birth, it was on the third day that the jaundice was noticed and we immediately went to the hospital. (FGD/female caregiver/Kano)

Treatment practices

The caregivers reported using both traditional and conventional medical treatments for NNJ. They describe using unorthodox remedies such as papaya leaves, Prussian blue, glucose water and exposure to unfiltered sunlight.

I heard that when they give birth to a child whose body suddenly becomes yellowish, he is said to have jaundice and that to treat it, the child can be brought out and kept under the sun. 'Bula' [*Prussian Blue, a blue substance used during laundry*] can also be used [*to cure jaundice*]. (FGD/female caregiver/Zaria)

Some also mentioned biomedical treatments, including phototherapy and exchange transfusions. "Once jaundice is noticed in a child, the doctor's directive should be followed: phototherapy, blood transfusion, and medications" (FGD/female caregiver/Jos).

Several caregivers reported receiving non-evidence-based recommendations from health workers or relatives, resulting in reliance on home remedies before seeking professional care.

I noticed the baby became jaundiced on the 3rd day of life and was advised to commence glucose and put him under sunlight. (FGD/female caregiver/Zaria)

Another caregiver stated,

Within the 3 days we were giving her glucose and other medications her father bought for us. (FGD/female caregiver/Jos)

Delays in accessing care

Caregivers reported delays in seeking and accessing care for neonates with jaundice that occurred at multiple levels: individual, misinterpretation and reliance on home remedies.

Interpersonal: inadequate counselling and family decision-making. Many caregivers initially did not recognise jaundice as a potentially life-threatening condition. The yellow discolouration of the skin was commonly interpreted as tiredness or a normal newborn condition. One caregiver explained, “We did not seek treatment for jaundice as we were told she was tired from birthing. We went to the hospital the next morning immediately when she began to be weak” (FGD/female caregiver/Zaria). Another caregiver stated,

She was healthy at birth. On the fifth day, her eyes became yellowish. My aunt told me to expose her to sunlight. She was very sleepy and we were busy getting ready for the naming ceremony. After the naming ceremony we took her to AKTH and she had EBT [exchange blood transfusion]. (FGD/female caregiver/Kano)

Similarly, some respondents indicated that health workers did not convey a sense of urgency, leading them to underestimate the seriousness of the illness. This reflects a lack of awareness of the dangers of NNJ and inadequate health education at delivery or postnatal care points.

Caregivers’ ability to seek timely care was also affected by personal or family crises. In some cases, bereavement, exhaustion, or lack of spousal support delayed health-seeking decisions. One respondent stated, “About the same time I was discharged, I lost my mother. So we could not go to the hospital till the 5th day” (FGD/female caregiver/Zaria).

Health system delays occurred even when caregivers sought medical attention promptly and these delays persisted due to inefficient referral processes, lack of bed space and resource limitations at health facilities. The following quotes exemplify this point: “We were referred after she became ill and we were only able to get to Shika the next day due to lack of bed”. “We went to BDT Hospital but could not be admitted there as their unit was under renovation. We were referred to another Hospital where he was commenced on phototherapy and oxygen after a six hour waiting” (FGD/female caregiver/Zaria); “We went to a Specialist Hospital, there was no bed. We went to another Hospital, no bed. We went to AKTH, no bed. We went home and gave traditional medicine” (FGD/female caregiver/Kano); Home Management “My child was yellowish at a week of age. We stayed at home

and gave traditional medicine. I thought all was well not until 5 months later, when my neighbor said to me ‘this child is not well’ (FGD/female caregiver/Kano).

Caregivers’ experience caring for children with KSD

Caregivers revealed that caring for children with KSD is tedious, time-consuming, costly and an energy-usurping exercise. A caregiver whose 7-year-old child has KSD expressed his experience as follows:

He is seven years. He is struggling to walk, but I can send him to pick things for me. But the problem is that you cannot leave him alone. I remembered the day I left him alone; he opened my toolbox, picked a razor blade and cut his hands. It was very painful! I was disturbed. Since then, we do not leave him alone. It’s either me or my wife. We have to watch him. If he wants to defecate, he will hold me and start dragging me – that’s a sign. He seems to understand things well but has difficulty in verbal expression. (FGD/male caregiver/Jos)

Stigma, community beliefs and misconceptions

Stigma was pervasive and ranged from jeering, exclusion from activities including education, to viewing the affected as not human, based on superstitious beliefs, as exemplified by these quotes:

That one is certain. I think there is nobody here that will say she has never been stigmatized. Even in my own church, when the child was much younger, then I was able to carry him along. When we went to church and sat close to a pregnant woman (one particular), she would leave the place and go to another place as if it was contagious. (FGD/female caregiver/Jos)

Another caregiver said,

Sometimes they say, this child is an ‘Ogbanje’, or a marine spirit. I had several occasions where I was told that these kinds of children are not normal, that if you keep them close to the river they will turn to snakes and crawl to the river. That I should go and meet a native doctor for medicine, that in the night, they will turn into one old man, they will walk, eat all your food, they will wash. They will do everything for themselves. That in the morning, when you wake up, you will see them as babies. One was even giving me advice that if I can come to where they can take away this burden from me, that I will be seeing the baby as a snake, she will enter the water and will not come back again. (FGD/female caregiver/Jos)

For further examples of stigma, community beliefs and misconceptions, see online supplemental table 1.

Perspectives from older children/young adults who could read and write

One young adult participant from Kano described his experience living with mild kernicterus. He managed to achieve significant milestones despite living with KSD. Reflecting on his journey, he shares “My Life as a Mild Kernicterus Victim”.

I attended SS for kindergarten, then moved to Malaysia for my primary education. We returned to Nigeria in 2015, and I continued my education at a Preparatory (school), earning my elementary and middle school certificates in 2018. Now, I am in Senior Secondary School 2 (Sciences) at an Academy, equivalent to grade 11.

He candidly discusses his challenges, which include difficulties with fine motor skills such as ironing clothes, playing soccer and writing. He also mentions challenges with making friends and maintaining concentration. Despite these hurdles, he participated in a National Mathematics Competition, ranking 92nd out of 1304 candidates—a testament to his cognitive abilities and determination which he attributes to his resilience and the unwavering support of his parents.

“My dad always challenges me to do what other normal children are capable of doing. My mom, with her love and care, has helped me discover my strengths, especially in computer skills”. YA dreams of becoming a software developer, aspiring to create revolutionary applications like Facebook and Adobe Acrobat. “My ambition is to become a software developer and create an app that can revolutionize the world”.

Additionally, we obtained the perspectives from three other young adults living with KSD who could read and write. We gained insight into their educational experiences and aspirations, challenges with mobility, communication and social stigma (online supplemental table 2).

DISCUSSION

To our knowledge, this is the first qualitative study on the experiences of both caregivers and children/individuals affected by KSD. The results of this study suggest that caregivers have a mix of both evidence-based knowledge and local cultural practices that together lead to an incomplete understanding of the disease, its treatment and sequelae. Caregivers were generally aware of NNJ, but their knowledge of causes was often flawed or, at best, incomplete. This has been documented in numerous papers.^{26–29} Common misunderstandings regarding perinatal factors that may prevent or exacerbate the development of jaundice, such as ‘eating chicken meat or oily food’ and non-compliance with antenatal care drugs. Cultural beliefs, knowledge, sources of information and practices play a significant role in shaping perceptions and should be considered when designing educational programmes, highlighting the need for robust health

communication strategies. Our goals should promote the culture of protecting the brains of newborns as we strive to save their lives.

Caregivers had a mix of medical and traditional treatment approaches to NNJ, including evidence-based phototherapy and exchange blood transfusion (EBT), while also relying on unproven remedies, such as glucose water, sugarcane, herbal treatments and early morning (unfiltered) sunlight. Many of these treatments were ineffective and sometimes harmful, especially in newborns with G6PDd. While appropriately filtered sunlight has been shown to be safe and effective for treating NNJ, early morning sunlight is ineffective, and mid-day can be fraught with hyperthermia, sunburn, etc.^{30–32 33} This misinformation indicates a need for targeted public health information about NNJ, which can be achieved through engaging community leaders, traditional birth attendants (TBAs) and other traditional healers.

Delays in seeking healthcare also occurred due to systemic issues, including a lack of beds. Personal choices (ie, prioritising traditional remedies such as papaya, Prussian blue ‘Bula’, glucose water, etc) were also problematic. Promoting awareness combined with maternal education (antenatally with postpartum reinforcement), screening, accurate, timely diagnosis and effective treatment of severe NNJ underscores the urgency of improving the capacity of primary healthcare centres to appropriately address these issues.^{5 32 34} Trained TBAs, community health extension workers and all healthcare providers are essential to this package.

Caring for children with KSD was described as emotionally, physically and financially draining for most of these vulnerable families living in a low-resource environment. Living with children with KSD made marriages and family interactions more challenging. Caregivers also shared experiences of managing developmental delays, nutritional challenges and the social stigma associated with KSD. Huge challenges in daily living activities, which are both life-altering and exhaustive, including difficulties in sitting, walking, toileting and expressing their needs, highlight the need for psychosocial and financial support for families. This finding is similar to what was reported from within Nigeria and other parts of the world from caregivers of children with CP.^{35–37}

All of these challenges occur in an environment that is almost exclusively not handicapped accessible and without the assistive technologies available in HICs. Developing low-cost technologies to assist these individuals will help alleviate some of the burden of care for individuals while increasing their ability to attain their goals and contribute to society. Community-based support networks, counselling services, financial and material support, along with respite care for caregivers, could mitigate stress, improve coping mechanisms and enable better caregiving.

All caregivers in this study acknowledged stigma. Their experiences ranged from overt actions, such as people avoiding proximity to their children due

to misconceptions of contagion, to subtle yet equally damaging behaviours, such as whispers and cruel comments. Stigma is rooted in ignorance and cultural misconceptions about disability. The emotional toll on caregivers, who often endure judgement, mockery and exclusion were highlighted in these FGDs. Male caregivers, in particular, shared experiences of being taunted by peers and even family members, highlighting the need for targeted interventions which challenge gender norms and reduce stigma for male caregivers. Public health education is critical to correcting such misconceptions. The interplay between stigma and cultural, religious and socioeconomic contexts shapes caregivers' experiences, how stigma manifests and coping mechanisms. Multi-sectoral collaboration involving healthcare providers, educators, policymakers, community and religious leaders can ensure a holistic approach with tailored interventions that account for local beliefs and practices are critical to combating stigma effectively.

As noted, before one caregiver said, "my child is an 'Ogbanje', a spirit ... that if I keep him close to the river, he will turn into a snake and crawl away." This haunting belief is not a relic of the distant past in Nigeria but a tragic reality confronting babies born today, babies whose devastating brain injuries were easily preventable. These caregivers are pressured to seek harmful traditional remedies or even consider abandonment to 'relieve the burden'. Some community members suggested killing these children because they are 'not human'. They have also viewed them as 'not normal' and as contributing much less than other family members. These beliefs not only stigmatise the children but also expose them to dangers, including abuse and isolate their caregivers. Collaboration with traditional and religious leaders, who wield influence in such communities, can help counter these narratives. Empowering leaders along with community sensitisation efforts to spread accurate health information on NNJ causing KSD, not supernatural powers, would promote acceptance and help alleviate stigma, reduce the danger to children and their families and combat the injustice they face.

However, while all caregivers faced stigma, some also reported sympathetic responses, such as assistance from strangers. Positive stories of community support should be amplified to inspire empathy and normalise caregiving, potentially creating broader networks of support for caregivers and their children.

Many children have challenges accessing basic education because of stigma, lack of qualified teachers and facilities for their special educational needs. Denial of admission to schools and/or caregivers being jeered by friends and family highlight systemic and interpersonal discrimination while limiting opportunities for children with KSD. Education plays a central role in the lives of all four individuals in the case study. Lack of concern for these children demonstrated by underfunding for special needs education at all levels of government makes the experiences of persons living with KSD in Nigeria worse than in HICs.³⁸ Inclusive schools and tailored learning approaches, including extended time

for assignments and physical assistance, helped overcome academic barriers and are important for empowering individuals with KSD. Raising awareness and advocacy about the rights of people with disabilities and implementing anti-discrimination laws is critical to and can foster a more inclusive society. Policies providing teacher training on disability inclusion are also essential for equitable learning experiences in LMICs.

The stories of the individuals with KSD demonstrate the challenges of living with this condition but highlight their resilience and ambition. Their aspirations included becoming a software developer, programmer and a pilot, aiming to defy societal limitations, serving as role models and demonstrating optimism and the potential of individuals with KSD when provided with supportive environments. Their brilliance is trapped-not by disability but by a world unwilling to see them as they are and to build physical and societal ramps they need to thrive. Their determination to excel in academics, acquire skills and contribute meaningfully to society is a testament to their inner strength. Advocacy efforts should focus on highlighting these success stories to challenge stereotypes and inspire others with similar conditions. Leveraging the voices of individuals can humanise disability and challenge discriminatory attitudes. Advocacy for affordable and accessible assistive devices is crucial to addressing these barriers. As with their caregivers, psychological support, such as counselling services, can help individuals process and cope with emotional challenges of living with KSD. Building peer support groups for these individuals can provide a sense of community and shared understanding. The participants advocate for a more inclusive and compassionate society, challenging the community to recognise the worth of individuals with disabilities, decrease stigma, amplify their voices and drive societal change. Collaboration with advocacy groups can help create platforms for these individuals to share their stories and influence policies.

Some caregivers know that their children are aware of what is going on and are of normal intelligence but unable to communicate effectively. This has been documented globally and is not limited to LMICs, as the USA branch of Parents of Infants and Children with Kernicterus (PICK) testifies. Sue Sheridan and a small group of mothers founded PICK in 2000 in response to the devastation she/they saw from this disease, coupled with the failure of some in the medical community to recognise the burden of this disease.³⁹ In a powerful and moving video done by eight of these older children/young adults from HICs, they also tell us of a world that 'looks at them without seeing'.¹⁷ Our children/adults in Nigeria are generally trapped in both a body and a society that is not working to protect and empower them to be all they have the potential and desire to be. Nevertheless, as stated powerfully in the words of one of our young adults, "I want to convince everyone that has been affected by kernicterus that no matter the obstacles, you can achieve all your dreams".

Today, that same statistical indifference perpetuates an even more deadly crisis in Nigeria, where

the lives of children are still treated as less valuable. Imagine if every month, a commercial airliner crashed, leaving dozens of children with lifelong disabilities. The global outcry would be urgent and loud, demanding investigations and systemic change. KSD is that ongoing catastrophe, yet because its victims are predominantly poor and living in LMICs, the world's gaze turns away. We tolerate for them what we would never accept for ourselves. But let us be clear: there is no 'acceptable' prevalence of KSD, only a brutal indictment of a dehumanising system. Our children mirror our collective conscience. They challenge us to stop making excuses and start raising our voices in a roar of outrage. This medical travesty must end now.

Global health systems, including those in Nigeria, must make the eradication of KSD a top priority. As pointed out by many within and outside Nigeria, KSD is preventable but not curable with limited treatments.¹⁻⁵ Data demonstrate that NNJ is the leading cause of hospitalisations in many studies in Nigeria and a leading cause of neonatal death in multiple studies in Nigeria.^{4-9 40} Additionally, it is second only to hypoxic ischaemic encephalopathy as a cause for CP, and a leading cause of deafness, which can be identified in the neonate with brainstem auditory evoked response testing.⁴¹⁻⁴⁵ Centring human stories can amplify the statistics to help drive more conversation, policy and guidelines.

This study highlights the significant challenges faced by caregivers and individuals with KSD, as well as their resilience in overcoming these obstacles. These findings also emphasise the need to educate families and society on how to prevent and treat severe NNJ/hyperbilirubinaemia. Acknowledging the challenges of living with KSD for these families and their children/adults, the need to 'see them' as who they are and recognising their potential to contribute to the Nigerian economy/prosperity, while advocating for them at all levels of society, is critical.

Limitations

Our FGDs, though multicentred, were limited to Northern Nigeria and therefore might not be generalisable to other LMICs. Additionally, the number of children/young adults is small and limited to participants who were easy to contact and well known to our group, which could have introduced bias into our analysis. We have yet to learn and understand the personal impact on the families whose newborns and children did not survive to teenage or adult life.

CONCLUSION

Kernicterus/KSD must be recognised as a Nigerian and worldwide public health emergency anywhere the risk for severe NNJ is high and resources to screen, diagnose and treat are limited. Minimal educational tools and the lack of a supportive environment make ABE/KSD more prevalent and challenging to live with. Small fixes or incremental

progress must be challenged by a pivotal aviation-safety health-societal agenda that views each KSD equivalent to an 'air crash'. Ours is a demand for the dismantling of systemic attitudes and taboos that perpetuate this tragedy. Let us invest in robust systems-based health initiatives rather than 'band-aid' interventions. Let us value and honour these human-life stories, not drowned in statistics, but centre each family's voice to transform every conversation into health policy- and patient-safety guidelines. Let us insist that these survivors, who were once healthy newborns, are no longer victimised but accorded the same opportunities afforded to other unaffected children. Let us protect all Nigerian newborns, making it possible for each child to achieve their dreams, fulfil their ambitions and reach their maximum potential.

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